

Computer Networking: HTTP and Web Requests

Short technical reference for testing PDF summarization and multi-step questions.

1. HTTP in Brief

HTTP is an application-layer protocol used for communication between clients and servers. A client sends a request, and a server normally returns a response. Web browsers, mobile applications, APIs, and many other systems use HTTP-based communication.

2. Common HTTP Methods

Method	Typical purpose
GET	Retrieve a representation of a resource.
POST	Submit data for processing or create a subordinate resource.
PUT	Create or replace a resource at a specified location.
PATCH	Partially modify an existing resource.
DELETE	Request removal of a resource.

3. Status Codes

Status codes communicate the result of a request. The 2xx class generally indicates success, 3xx indicates redirection, 4xx indicates a client-side problem such as an invalid request or missing authorization, and 5xx indicates a server-side failure.

4. Headers and Body

HTTP headers carry metadata such as content type, caching instructions, authorization information, and accepted response formats. The message body can contain data such as JSON, HTML, images, or other representations. Whether a request or response has a body depends on the method and the specific interaction.

5. Security Note

HTTPS protects HTTP traffic by using TLS. Encryption helps prevent unauthorized parties from reading or modifying traffic in transit, while certificates help clients authenticate the server's identity.